

CSE548 Project 1: Packet Filter Firewall (iptables) Project

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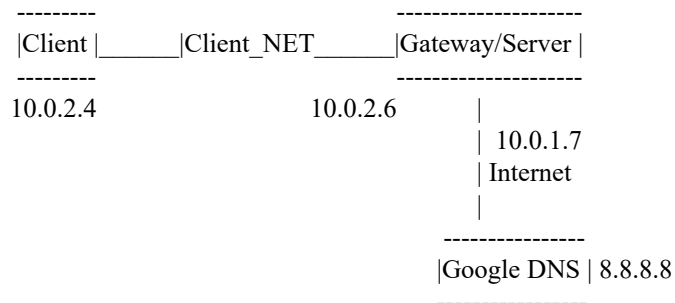
I. PROJECT OVERVIEW

In the project we set up an iptables firewall on a Gateway / Server that follows conditions determined by the following requirements:

1. A client can send pings to 8.8.8.8 though a Gateway / Server
2. A client can access a webpage residing on the Gateway / Server
3. A client can not access any IP addresses besides 8.8.8.8
4. The client can not ping the Gateway / Server
5. The server can not perform pings to localhost
6. The server can not perform pings to the client
7. The server can not perform pings to 8.8.8.8
8. The server provides Network Address Translation for the client so that packets from the client appear to come from the Gateway / Server.

II. NETWORK SETUP

Lab Network Topology

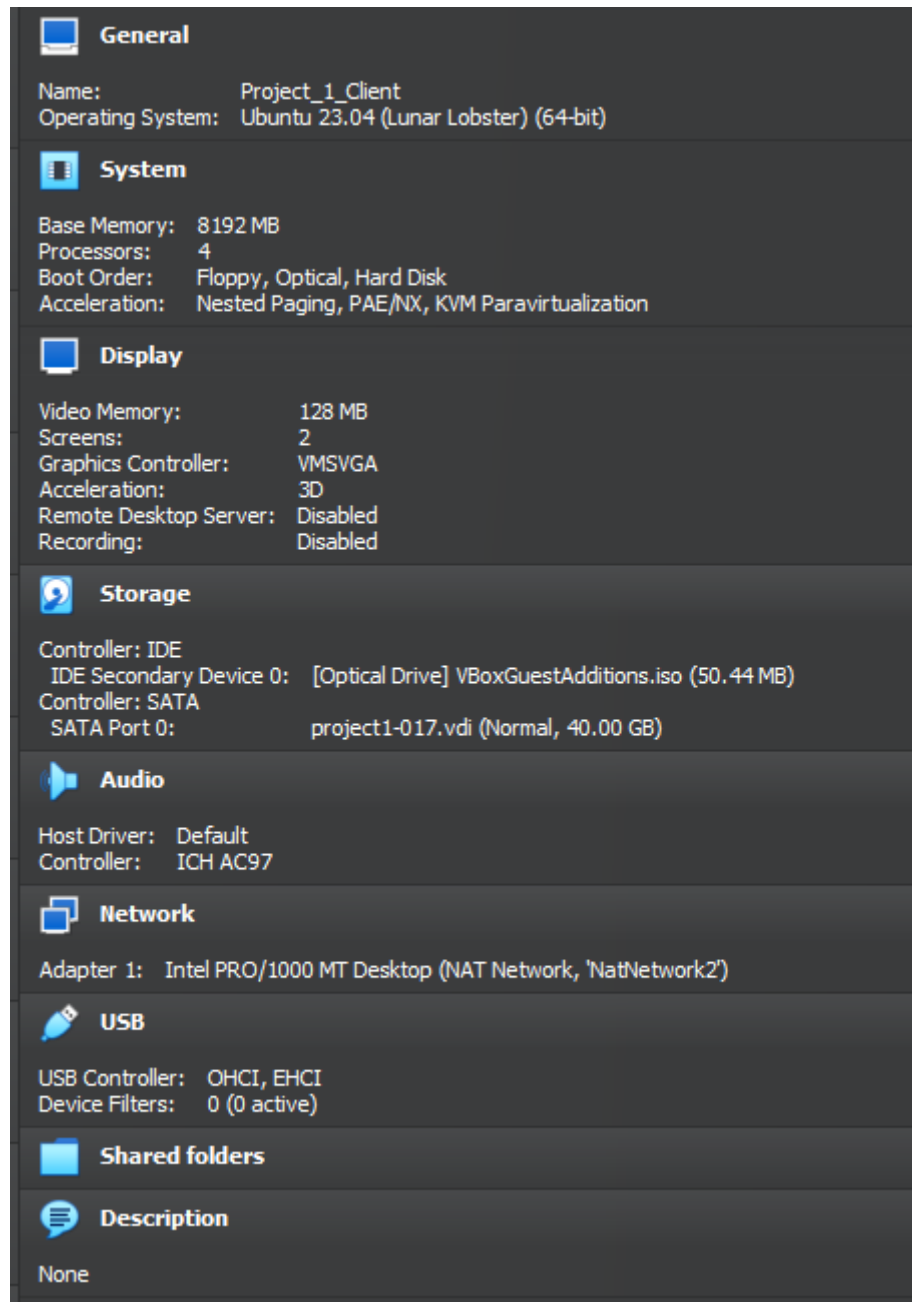


III. SOFTWARE

The software used for this project was Virtualbox for acting as the hypervisor for the Client and Gateway / Server virtual machines. Both virtual machines were based on the Ubuntu 18.04 Linux operating system.

IV. PROJECT DESCRIPTION

The virtual machines were set up in Virtualbox as follows:
Client:



The image shows the settings window for a virtual machine named 'Project_1_Client'. The window is divided into several tabs, each with a corresponding icon: General (monitor), System (circuit board), Display (monitor), Storage (floppy disk), Audio (speaker), Network (network card), USB (USB plug), Shared folders (folder), and Description (speech bubble). The 'General' tab is currently selected. The settings are as follows:

Section	Setting	Value
General	Name:	Project_1_Client
	Operating System:	Ubuntu 23.04 (Lunar Lobster) (64-bit)
System	Base Memory:	8192 MB
	Processors:	4
	Boot Order:	Floppy, Optical, Hard Disk
	Acceleration:	Nested Paging, PAE/NX, KVM Paravirtualization
Display	Video Memory:	128 MB
	Screens:	2
	Graphics Controller:	VMSVGA
	Acceleration:	3D
	Remote Desktop Server:	Disabled
	Recording:	Disabled
Storage	Controller:	IDE
	IDE Secondary Device 0:	[Optical Drive] VBoxGuestAdditions.iso (50.44 MB)
	Controller:	SATA
Storage	SATA Port 0:	project1-017.vdi (Normal, 40.00 GB)
	Audio	
Host Driver:		Default
Controller:		ICH AC97
Network		
Adapter 1:		Intel PRO/1000 MT Desktop (NAT Network, 'NatNetwork2')
USB		
USB Controller:		OHCI, EHCI
Device Filters:		0 (0 active)
Shared folders		
Description		
None		

Server:

 **General**

Name: Project_1_Server
Operating System: Ubuntu 23.04 (Lunar Lobster) (64-bit)

 **System**

Base Memory: 8192 MB
Processors: 4
Boot Order: Floppy, Optical, Hard Disk
Acceleration: Nested Paging, PAE/NX, KVM Paravirtualization

 **Display**

Video Memory: 128 MB
Graphics Controller: VMSVGA
Acceleration: 3D
Remote Desktop Server: Disabled
Recording: Disabled

 **Storage**

Controller: IDE
IDE Secondary Device 0: [Optical Drive] Empty
Controller: SATA
SATA Port 0: Project_1_Server-disk1.vdi (Normal, 40.00 GB)

 **Audio**

Host Driver: Default
Controller: ICH AC97

 **Network**

Adapter 1: Intel PRO/1000 MT Desktop (NAT Network, 'NatNetwork1')
Adapter 2: Intel PRO/1000 MT Desktop (NAT Network, 'NatNetwork2')

 **USB**

USB Controller: OHCI, EHCI
Device Filters: 0 (0 active)

 **Shared folders**

 **Description**

None

The client's routing table was set up as follows:

```
ubuntu@client:~$ route
Kernel IP routing table
Destination    Gateway         Genmask         Flags Metric Ref    Use Iface
default        _gateway        0.0.0.0         UG    0      0      0 enp0s3
default        _gateway        0.0.0.0         UG    20100  0      0 enp0s3
10.0.2.0       0.0.0.0         255.255.255.0   U     100    0      0 enp0s3
link-local     0.0.0.0         255.255.0.0     U     1000   0      0 enp0s3
ubuntu@client:~$ route -n
Kernel IP routing table
Destination    Gateway         Genmask         Flags Metric Ref    Use Iface
0.0.0.0        10.0.2.6        0.0.0.0         UG    0      0      0 enp0s3
0.0.0.0        10.0.2.1        0.0.0.0         UG    20100  0      0 enp0s3
10.0.2.0       0.0.0.0         255.255.255.0   U     100    0      0 enp0s3
169.254.0.0    0.0.0.0         255.255.0.0     U     1000   0      0 enp0s3
ubuntu@client:~$
```

The client's network setup:

```
ubuntu@client:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 10.0.2.4  netmask 255.255.255.0  broadcast 10.0.2.255
    inet6 fe80::72d2:25ac:dc72:f372  prefixlen 64  scopeid 0x20<link>
    ether 08:00:27:62:f5:9c  txqueuelen 1000  (Ethernet)
    RX packets 777  bytes 355720 (355.7 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 285571  bytes 19350942 (19.3 MB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0x10<host>
    loop txqueuelen 1000  (Local Loopback)
    RX packets 652  bytes 66332 (66.3 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 652  bytes 66332 (66.3 KB)
    TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

ubuntu@client:~$
```

The Server's network setup:

```
ubuntu@server:~/Documents/Project1$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.1.7 netmask 255.255.255.0 broadcast 10.0.1.255
    inet6 fe80::bdfa:6c2d:f264:526 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:a2:80:c8 txqueuelen 1000 (Ethernet)
    RX packets 543852 bytes 801984713 (801.9 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 188698 bytes 11451405 (11.4 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

enp0s8: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.6 netmask 255.255.255.0 broadcast 10.0.2.255
    inet6 fe80::fd23:5686:215b:aa2a prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:dd:6d:18 txqueuelen 1000 (Ethernet)
    RX packets 285246 bytes 19310918 (19.3 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 494 bytes 48448 (48.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 417 bytes 44914 (44.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 417 bytes 44914 (44.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ubuntu@server:~/Documents/Project1$
```

The Server's iptables rules were set up as follows:

```
ubuntu@server:~/Documents/Project1$ sudo iptables -L
[sudo] password for ubuntu:
Chain INPUT (policy DROP)
target     prot opt source                destination
DROP       icmp -- localhost            anywhere
DROP       icmp -- 10.0.2.4              server
ACCEPT     tcp  -- 10.0.2.4              anywhere        tcp dpt:http
ACCEPT     icmp -- anywhere             anywhere

Chain FORWARD (policy DROP)
target     prot opt source                destination
ACCEPT     icmp -- 8.8.8.8              anywhere
ACCEPT     all  -- 10.0.2.4              8.8.8.8
ACCEPT     tcp  -- anywhere             anywhere

Chain OUTPUT (policy DROP)
target     prot opt source                destination
DROP       icmp -- anywhere            anywhere
ACCEPT     tcp  -- anywhere            10.0.2.4        tcp spt:http
ubuntu@server:~/Documents/Project1$
```

The Server's Network Address Translations (NAT) was setup as follows:

```
ubuntu@server:~/Documents/Project1$ sudo iptables -t nat -L POSTROUTING
[sudo] password for ubuntu:
Chain POSTROUTING (policy ACCEPT)
target     prot opt source                destination
MASQUERADE all  -- anywhere            8.8.8.8
MASQUERADE all  -- anywhere            anywhere
ubuntu@server:~/Documents/Project1$ s
```

A Webserver was setup on the Gateway / Server with the following web content:

```
ubuntu@server:~/Documents/Project1$ cat /var/www/html/index.html
Welcome to Demo and Test!
ubuntu@server:~/Documents/Project1$
```

The webserver was activated and the status is as follows:

```
ubuntu@server:~/Documents/Project1$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Drop-In: /lib/systemd/system/apache2.service.d
            └─apache2-systemd.conf
   Active: active (running) since Sun 2024-10-27 13:30:06 MST; 1h 26min ago
     Process: 5035 ExecStop=/usr/sbin/apachectl stop (code=exited, status=0/SUCCESS)
     Process: 5040 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 5044 (apache2)
    Tasks: 55 (limit: 4915)
   CGroup: /system.slice/apache2.service
           └─5044 /usr/sbin/apache2 -k start
             └─5045 /usr/sbin/apache2 -k start
               └─5046 /usr/sbin/apache2 -k start

Oct 27 13:30:06 server systemd[1]: Starting The Apache HTTP Server...
Oct 27 13:30:06 server apachectl[5040]: AH00558: apache2: Could not reliably determine the server's fully qualified domain
Oct 27 13:30:06 server systemd[1]: Started The Apache HTTP Server.
```

When the client uses curl to access the website on the server, the result is as follows:

```
ubuntu@client:/mnt$ curl 10.0.2.6
Welcome to Demo and Test!
ubuntu@client:/mnt$
```

In order to test proper functioning of the firewall, the following port scans were performed:

Client TCP port scan of Server

```
$ sudo nmap -sT -p- 10.0.2.x % x is the value of your  
Gateway/Server VM's IP address
```

```
ubuntu@client:/mnt$ sudo nmap -vv -sT -p- 10.0.2.6

Starting Nmap 7.60 ( https://nmap.org ) at 2024-10-27 14:28 MST
Initiating ARP Ping Scan at 14:28
Scanning 10.0.2.6 [1 port]
Completed ARP Ping Scan at 14:28, 0.22s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 14:28
Completed Parallel DNS resolution of 1 host. at 14:28, 13.01s elapsed
Initiating Connect Scan at 14:28
Scanning 10.0.2.6 [65535 ports]
Discovered open port 80/tcp on 10.0.2.6
Connect Scan Timing: About 19.14% done; ETC: 14:31 (0:02:11 remaining)
Connect Scan Timing: About 45.40% done; ETC: 14:31 (0:01:13 remaining)
Completed Connect Scan at 14:30, 108.54s elapsed (65535 total ports)
Nmap scan report for 10.0.2.6
Host is up, received arp-response (0.0033s latency).
Scanned at 2024-10-27 14:28:34 MST for 122s
Not shown: 65534 filtered ports
Reason: 65534 no-responses
PORT      STATE SERVICE REASON
80/tcp    open  http   syn-ack
MAC Address: 08:00:27:DD:6D:18 (Oracle VirtualBox virtual NIC)

Read data files from: /usr/bin/../share/nmap
Nmap done: 1 IP address (1 host up) scanned in 121.83 seconds
Raw packets sent: 1 (28B) | Rcvd: 1 (28B)
ubuntu@client:/mnt$
```

Client UDP port scan of Server

```
$ ubuntu@client:/mnt$ sudo nmap -vv -sU -p- 10.0.2.6

Starting Nmap 7.60 ( https://nmap.org ) at 2024-10-27 14:32 MST
Initiating ARP Ping Scan at 14:32
Scanning 10.0.2.6 [1 port]
Completed ARP Ping Scan at 14:32, 0.23s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 14:32
Completed Parallel DNS resolution of 1 host. at 14:32, 13.01s elapsed
Initiating UDP Scan at 14:32
Scanning 10.0.2.6 [65535 ports]
UDP Scan Timing: About 2.18% done; ETC: 14:56 (0:23:10 remaining)
UDP Scan Timing: About 6.47% done; ETC: 14:55 (0:21:56 remaining)
UDP Scan Timing: About 12.15% done; ETC: 14:55 (0:20:43 remaining)
UDP Scan Timing: About 17.26% done; ETC: 14:55 (0:19:30 remaining)
UDP Scan Timing: About 22.01% done; ETC: 14:55 (0:18:19 remaining)
UDP Scan Timing: About 26.78% done; ETC: 14:55 (0:17:08 remaining)
UDP Scan Timing: About 32.62% done; ETC: 14:56 (0:15:56 remaining)
UDP Scan Timing: About 37.46% done; ETC: 14:55 (0:14:43 remaining)
UDP Scan Timing: About 42.45% done; ETC: 14:55 (0:13:31 remaining)
UDP Scan Timing: About 47.50% done; ETC: 14:55 (0:12:17 remaining)
UDP Scan Timing: About 52.66% done; ETC: 14:55 (0:11:04 remaining)
UDP Scan Timing: About 57.64% done; ETC: 14:55 (0:09:54 remaining)
UDP Scan Timing: About 62.78% done; ETC: 14:55 (0:08:42 remaining)
UDP Scan Timing: About 67.89% done; ETC: 14:55 (0:07:30 remaining)
UDP Scan Timing: About 73.00% done; ETC: 14:55 (0:06:19 remaining)
UDP Scan Timing: About 78.04% done; ETC: 14:55 (0:05:08 remaining)
UDP Scan Timing: About 83.07% done; ETC: 14:55 (0:03:57 remaining)
UDP Scan Timing: About 88.13% done; ETC: 14:55 (0:02:47 remaining)
UDP Scan Timing: About 93.32% done; ETC: 14:55 (0:01:34 remaining)
Completed UDP Scan at 14:55, 1401.93s elapsed (65535 total ports)
Nmap scan report for 10.0.2.6
Host is up, received arp-response (0.0010s latency).
All 65535 scanned ports on 10.0.2.6 are open|filtered because of 65535 no-responses
MAC Address: 08:00:27:DD:6D:18 (Oracle VirtualBox virtual NIC)

Read data files from: /usr/bin/./share/nmap
Nmap done: 1 IP address (1 host up) scanned in 1415.26 seconds
Raw packets sent: 131071 (3.674MB) | Rcvd: 32 (2.702KB)
ubuntu@client:/mnt$
```

Ping Tests to Verify Firewall Requirements

Client ping to internet address 8.8.8.8

```
$ ping 8.8.8.8 % This should be working
```

```
ubuntu@client:/mnt$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=114 time=20.6 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=114 time=20.3 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=114 time=24.4 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=114 time=21.3 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=114 time=20.5 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=114 time=20.5 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=114 time=22.9 ms
^C
--- 8.8.8.8 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6038ms
rtt min/avg/max/mdev = 20.343/21.556/24.486/1.459 ms
ubuntu@client:/mnt$
```

Client ping to internet address 8.8.4.4

```
$ ping 8.8.4.4 % This should be not working, as you should drop
all traffic that is not required in the
requirement.
```

```
ubuntu@client:/mnt$ ping 8.8.4.4
PING 8.8.4.4 (8.8.4.4) 56(84) bytes of data.
^C
--- 8.8.4.4 ping statistics ---
9 packets transmitted, 0 received, 100% packet loss, time 8175ms
ubuntu@client:/mnt$
```

Client ping of Server

```
$ ping 10.0.2.x % x̄ is the value of your Gateway/Server VM's IP  
address, This should be not working
```

```
ubuntu@client:/mnt$ ping 10.0.2.6  
PING 10.0.2.6 (10.0.2.6) 56(84) bytes of data.  
^C  
--- 10.0.2.6 ping statistics ---  
27 packets transmitted, 0 received, 100% packet loss, time 26622ms  
  
ubuntu@client:/mnt$
```

Server ping to localhost

```
$ ping localhost % This should be not working
```

```
ubuntu@server:~/Documents/Project1$ ping localhost
PING localhost (127.0.0.1) 56(84) bytes of data.
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
^C
--- localhost ping statistics ---
5 packets transmitted, 0 received, 100% packet loss, time 4083ms

ubuntu@server:~/Documents/Project1$
```

Server ping to Client

```
$ ping 10.0.2.y % y is the value of your client's IP address,
this should be not working
```

```
ubuntu@server:~/Documents/Project1$ ping 10.0.2.4
PING 10.0.2.4 (10.0.2.4) 56(84) bytes of data.
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
^C
--- 10.0.2.4 ping statistics ---
9 packets transmitted, 0 received, 100% packet loss, time 8196ms

ubuntu@server:~/Documents/Project1$
```

V. CONCLUSION

A lesson that I learned was that it was worth taking the time to enter the iptables commands one at a time and checking the iptables chains using iptables -L. I also learned how to fix errors by using the --line-numbers switch and the -D switch to delete mistakes. Iptables chains can become complicated, so it is worth to flush the entire chains and start over when things stop working. I also learned that -A will append a rule to the end, while -I will insert a rule to the top. The order of the rules matters, so that if there is a DROP before an ACCEPT, then the packet will be dropped.